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Leveraging quantum computing business applications: a B-SLR approach and future research directions

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Abstract

Quantum computing (QC) advances will help solve high complexity challenges that classical system architectures cannot solve. This report examines the philosophical foundation for QC research using a PRISMA-compliant systematic literature review and bibliometric analysis. This review emphasizes business applications of QC, including optimization, cryptography and cybersecurity, machine learning, financial modeling and forecasting, supply chain management, healthcare and analytics, and materials design, showing how QC can improve prediction and forecast accuracy, data physicalness, large-scale optimization, and more.

The research will also identify execution constraints such hardware readiness, qubit capacity, development and operation expenses, personnel availability, and regulatory and commercial governance. Bibliometric data showed a focus on foundational quantum theoretical contributions and algorithms, with multidisciplinary and practical work emerging. Scaled algorithms, quantum-resistant standards, and any solutions industry nexus to enterprise are among the research opportunities. The literature evaluation will also identify ethical, policy, and sustainability issues related to quantum benefits.

The paper helps businesspeople understand QC's current and future opportunities, including a research-based assessment of where we've come, common themes in the literature, and strategic domains for scaling QC as a research agenda from laboratory experiments to enriching the service experience as commercial utility.

Keywords: Quantum computing; Business applications; Bibliometric-systematic literature review; PRISMA; NISQ; Quantum readiness; Technology adoption; Strategic management

1 Introduction

Quantum Computing is a significantly different paradigm from Classical Computing since it operates on Quantum Mechanical principles such as Superposition, Entanglement and Interference. Quantum Computers will solve some problems faster than Classical Computers can ever hope to, as confirmed by the foundational research done on Quantum Computing at Stanford University by Professor Simon Levinson [21]. The foundations of

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Quantum Computing were laid in the early 1980s with the work of Richard Feynman [10] and the invention of the concept of Quantum Simulation and later validated by major discoveries in the early 1990s with the invention of Shor's factoring algorithm [26] and Grover's search algorithm [13]. The theoretical underpinnings of Quantum Computing therefore firmly established Quantum Computing as a viable means for solving certain classes of problems.

The last decade has seen Quantum Computing grow from mostly theoretical research to a rapidly growing market in both technology and commercial domain. The rapid advancement of Quantum Computing has resulted from advances in the operations of Quantum Computers by hardware engineers, the amount of money that has been invested by both publicly and privately to support Quantum Computing development and the participation of many major tech companies in the Quantum Computing ecosystem [25]. The current stage of Quantum Computing is known as the NISQ (Noisy Intermediate-Scale Quantum) stage, which is comprised of Quantum Processors containing anywhere from about 50 physical Qubits up to about 1000 physical Qubits, which have certain limitations due to the high level of restrictions placed upon Quantum Processors by decoherence, gate errors, and limited depth of Quantum Circuits [3, 25]. Despite these limitations, the proliferation of cloud-based Quantum Computing services has decreased the barriers to entry into the Quantum Computing market and has allowed more people the opportunity to use Quantum Computers to develop experiments and to create a highly competitive arena within the international community for those seeking to gain a competitive advantage in Quantum Computing. The recent achievements of Google's Sycamore processor has showcased Quantum Supremacy and the recent advancements in logical qubit stability along with improvements in Quantum Error Correction, as demonstrated by the research of Arute et al. [2] and Acharya et al. [1], confirms considerable progress has been made toward achieving fault-tolerant quantum computing. But for large scale commercial applications of quantum computers using error corrected quantum systems, many years or decades still remain before they can be built. Thus near-term value for the time being will come primarily through use of hybrid quantum-classical configurations and application-specific algorithms that run within the limitations of devices in the Noisy Intermediate-Scale Quantum (NISQ) era.

The recent growth of interest from the business sector coupled with advancements in technology presents significant business opportunities for organizations throughout a variety of sectors. Organizations are exploring the use of quantum technology for its potential quantum-enabled advantages in areas such as optimization, simulation, machine learning and cryptography [14, 15]. From the financial sector's pursuit of using quantum technology for optimizing portfolio management and for analysing risk [8, 22] to those working within the pharmaceutical industry examining the use of quantum technology for enhancing molecular simulation and drug discovery efforts [4, 23], and logistics providers examining the potential for increased efficiency through routing and scheduling optimization [9] to public sector organizations preparing for the future implementation of anti-quantum cryptography [7, 20], quantum computing has the potential to transform how businesses will address their ever-changing marketplace competition through the creation of innovative new products and services.

While there is increasing interest in quantum computing from both the academic and business communities, the level of risk associated with transforming quantum comput-

ing into an expense for companies will be extremely high due to the many unknowns surrounding the development of quantum computing and the associated infrastructure changes needed to support the new systems and processes required to support their implementation. Organizations are currently faced with significant barriers to quantum computing, including underdevelopment in technology, lack of quantum skilled labour, challenges integrating classical information systems using quantum computing principles, lack of clear regulations, and uncertainty surrounding return on investment timing. In a period of rapid growth in hardware capability, vast amounts of capital coming from public and private investments in quantum technology, and government support for quantum technology as a fundamental building block for national strategic solutions, organizations are exposed to your business and operational risks by not having structured quality to support their investment decisions. Without high-quality research available to support their business investment decisions, an organization could potentially make too large of an investment in technology too early, misinterpret the timeframes to achieve quantum computing's practical applications, and/or make strategic plans to protect an organization from future disruption caused by quantum computing.

The literature on quantum computing has grown exponentially, yet it continues to be missing key components of information that are essential to enable sound strategic decision making and successful escalation of knowledge creation for business and management types of research. Literature reviews pertaining to quantum computing tend to focus on algorithms and their requirements [5, 16, 19] through computational complexity, specific algorithm families, or types of quantum computing hardware architecture. Although each of the reviews is highly regarded within its own respective discipline, organizational readiness for the adoption of quantum computing, the specific pathway for quantum implementation within individual sectors, and the means to create and deliver strategic value related to quantum computing is not addressed within these reviews. Additionally, the literature often characterizes quantum computing near-term, NISQ technology capabilities as equivalent to long-term fault tolerant quantum computing capabilities, thereby resulting in unrealistic expectations about how quickly quantum computing will be adopted and available to organizations [3, 17]. Furthermore, the current body of research on quantum application development continues to exist in fragmentation across business disciplines, where quantum computing studies in finance [22], healthcare [9], logistics [27], cybersecurity [20], and materials science.

This requires extensive research to determine how companies should plan and successfully implement Quantum Computing into their Business Model. Therefore, this study utilizes the methodology of quantitative Bibliometrics to carry out a systematic review of the literature relating to Quantum Computing and its potential uses across multiple industries, including Technology Readiness, Organisation Readiness and Strategic Value Creation.

2 Research objectives

- To systematically synthesize the literature on quantum computing business applications, identifying core application domains, technological requirements, and implementation characteristics.
- To map the intellectual structure, research evolution, influential contributions, and emerging themes within the quantum computing business applications literature using bibliometric analysis.

- To assess sector specific quantum readiness by explicitly distinguishing NISQ era feasibility from fault tolerant quantum requirements and identifying adoption barriers and indicative timelines.
- To develop an integrative framework and future research agenda to inform organizational strategy, policy design, and scholarly inquiry related to quantum computing adoption.

3 Research questions

RQ1: What foundational concepts and ideas form the basis of quantum computing; what major themes can be found; and how is quantum computing evolving within the field of it?

RQ2: How does quantum computing provide value to organizations today; what are the major technological developments, barriers to implementing quantum computers in business practices and factors that will determine the success or otherwise of quantum computing for companies?

RQ3: What will be the overall influence of quantum computing on Management, Organization, and Policy within today's businesses?

RQ4: Where are the most significant research gaps today based on current understanding of where quantum computing has been positioned, and where do we have opportunities to move forward with quantum technologies?

4 Theoretical framework

By taking a theoretical approach to conceptualizing quantum computing from both a technology perspective and organisational resource perspective, this study utilises complementary theoretical perspectives.

The Hybrid Classical Quantum integration framework represents a view of Quantum Computing as being integrated into the existing classical ecosystem, while also suggesting that quantum computing will be a cloud-based, complementary computational resource to existing systems through Quantum Computing as a Service for Hybrid Classical Quantum Development (QCSHQD), where the emphasis is on establishing abstraction layers and hybrid workflows to reduce barriers to entry through expertise and develop incremental value over time, under the current NISQ limitations. Additionally, the Quantum Parallelism Framework (QPF) emphasizes the need for the co-design of classical and quantum layers, in order to operationalize quantum parallelism in the context of industrial workflows.

Diffusion and Adoption theories can help explain the atypical diffusion characteristics of quantum computing. Particularly through Theories of Diffusion of Innovations and Crossing the Chasm, these theories identify the intrinsic uncertainty, delayed return on investment, and reliance on the network of partner organisations, and social and institutional legitimacy, in the early stages of adoption.

The Resource-Based View (RBV) provides a strategic lens through which to evaluate the potential for quantum computing to be a source of competitive advantage. The capabilities of Quantum Computing may provide value to certain classes of problems, are unique and difficult to duplicate due to levels of accessibility and levels of specialization/knowledge required for operating quantum computing and are also extremely expensive to replicate due to the infrastructure and knowledge required to run Quantum Computing. While

Quantum Computing exhibits various degrees and types of VRIN capabilities under the NISQ paradigm, it is likely that the benefits of early Quantum Computing will be transitory rather than sustained. By understanding the differences between exploratory investment and investment that units or organisations can generate strategic benefits (defensible), the RBV assists in differentiating exploratory investments from those that can produce a sustainable strategic advantage.

Cloud native technology perspectives extend this to provide an understanding of quantum computing in the context of a distributed service architecture. As such, the Quantum Function-as-a-Service (QFaaS) paradigm represents a model of quantum execution as an event-driven and modular architecture that is interchanged and integrated through heterogeneous Cloud-computing providers, while complying with existing DevOps and orchestration best practices.

Security oriented models position Quantum Computing as a paradigm shift from a computationally based model of security to a physics-based model. In particular, the concept of Quantum Key Distribution (QKD) is an example of this, in that it establishes a provably secure communication model by applying the no-cloning theorem in quantum computing, and by doing so, illustrates the implications to Cybersecurity Governance and Cybersecurity Infrastructure and Compliance Structures.

All of these perspectives frame Quantum Computing as an integrative socio-technical transformation that organisational impact is largely dependent upon the degree of alignment between the organisation's technological capability (technology readiness), organisational capability (technology infrastructure and skillsets), and organisational understanding of the strategic value of quantum computing (Return on Investment).

5 Methodology

This research applies a Bibliometric-Systematic Literature Review (B-SLR) framework to investigate procedures, structure, and business applications over the course of Quantum Computing research development. By integrating Bibliometric Analysis with a systematic literature review (guided by PRISMA), a dual objective of quantitative mapping of the intellectual landscape and qualitative synthesis of application-level insights is achieved with greater analytical rigor, transparency, and replicability.

5.1 Who conducted the search and screening?

The literature search and data collection activities of the identified records were completed by the authors using a structured, comprehensive and reproducible search protocol. The lead author performed the initial database search and refinement, as well as screening activities. The co-author reviewed the search strategy and screening decisions independently of the lead author prior to the collection of data in order to reduce selection bias and enhance reliability. Any discrepancies in the original inclusion decision were resolved through discussion and consensus, following established best practice procedures for systematic reviews [18, 24].

5.2 When were the data collected?

To ensure the review would address current advances in Quantum Computing and associated business uses, only articles published from 2015 to 2025 were included in the analysis. This period includes the beginning of what has been termed the "Noisy Intermediate-Scale

Quantum" (NISQ) era, the development of cloud quantum platforms, and the increasing interest in Quantum Computing for business and management applications.

5.3 Where were the data collected?

Data were collected exclusively from peer-reviewed journal articles contained within the Scopus database. Scopus was chosen due to its broad coverage of quality journals across a wide range of disciplines, including Business, Management, Economics, Decision Science, and Computer Science. Thus, Scopus provided an appropriate resource for data collection for the interdisciplinary nature of Quantum Computing application in business settings. In accordance with accepted methodological practices in management research, the articles were chosen over books, proceedings and reports because they represent a rigorous process of peer review allowing for a more reliable basis for future research by providing stronger theoretical foundation(s) as well as greater empirical consistency than all other types of publications.

5.4 How were the studies identified?

A Boolean search strategy was designed to capture the intersection of quantum computing and business applications. The primary search string used was:

"quantum" AND "computing" AND "business" AND "applications"

The initial search yielded 166 records. To ensure relevance and quality, the results were refined using predefined filters: document type (articles only), source type (journals), language (English), and publication stage (final). Subject area filters were applied to retain studies published within Business, Management and Accounting, Economics, Econometrics and Finance, Decision Sciences, and Computer Science.

5.5 What was included and excluded?

After the refinement process was completed, the rest of the records were evaluated with a multi-level approach. First, the titles and abstracts were read to determine if they were relevant to business or organizational contexts concerning quantum computing. Articles that were focussed on quantum physics as a science, hardware engineering, and purely technological algorithms without any relevance to a business were eliminated. After this first screening, all remaining studies then underwent a comprehensive review to determine if they matched the research objectives of this study. After completing all three stages of screening and refinement, 33 peer-reviewed journal articles met the criteria for inclusion and were included in the more in-depth analysis portion of this work.

5.6 Inclusion and exclusion criteria

Inclusion criteria: Peer-reviewed journal articles indexed in Scopus, published between 2015 and 2025, written in English with explicit focus on quantum computing in business, management, organizational, or applied economic contexts.

Exclusion criteria: Conference papers, books, book chapters, editorials, reports, and non-peer-reviewed material, articles focused exclusively on quantum physics or hardware without business relevance, studies published outside the defined time frame and non-English publications

5.7 Bibliometric analysis procedure

To conduct a bibliometric analysis, researchers used the final dataset to find the research evolution and thematic concentration within quantum computing business research as well as map out the intellectual structure related to this field. The Publication Trends, Citation Patterns, Leading Authors and Journals and Thematic Clusters were included in the Bibliometric Mapping based on multiple sources of information. Citation Indicators are used to indicate the level of impact of a given body of research; Co-Authorship and Co-Occurrence Keyword analyses were employed to support understanding of researcher collaboration patterns and emerging research themes. Bibliometric maps were developed using Scopus Index Metadata that is both Standard & Comprehensive for citation information and represents extensive Interdisciplinary coverage.

5.8 Systematic synthesis and data analysis

After the completion of the bibliometric analysis, a qualitative systematic review of all the articles included in the quantitative analysis was performed. Each article was evaluated for its relevant factors to quantum computing applications, implementation and strategic implications, and these factors were then synthesised qualitatively across different organisations and sectors. The qualitative synthesis of the study data answers the study's research questions and lists the barriers that prevent the application of quantum technology, along with the major themes that were repeated across all the different sectoral contexts. Therefore, barriers to quantum computing and through research gaps due to lack of applicability will be identified.

5.9 Why (final selection rationale and PRISMA compliance)

The Systematic review's credibility is entirely dependent upon both the quality and relevance of the studies included in it. As such, articles were retained only if they were a direct source of information regarding Quantum Computing as used in Business or Organizational settings. By focusing on this specific subject area, it provides clarity of concept and greater depth of analysis.

The stages of identifying, screening, eligibility, and inclusion are conducted according to the guidelines established by the PRISMA 2020 protocol. Figure 1 below, which is a PRISMA flow diagram, shows how the selection process was conducted, providing transparency, methodological rigor, and enabling replication of this review.

6 Results

6.1 Bibliometric analysis

Table 1 provides a summary of annual scientific output on quantum computing from 2004 to 2025, as well as some observations about the changes in research activity in this area. From 2004-2012, the scientific output was comparatively lower than in more recent years, and we attribute this to the developmental stage of quantum computing that was primarily theoretical at this time.

The research output increased somewhat in 2013-2014, which we attribute to advancements in quantum hardware and an increase in research community interest across the globe. After a slight plateau from 2015-2019, the annual series of publication engagements greatly accelerated starting in 2020. This aligns with various high-level events in the technology, including reported instances of quantum supremacy and the commercialization

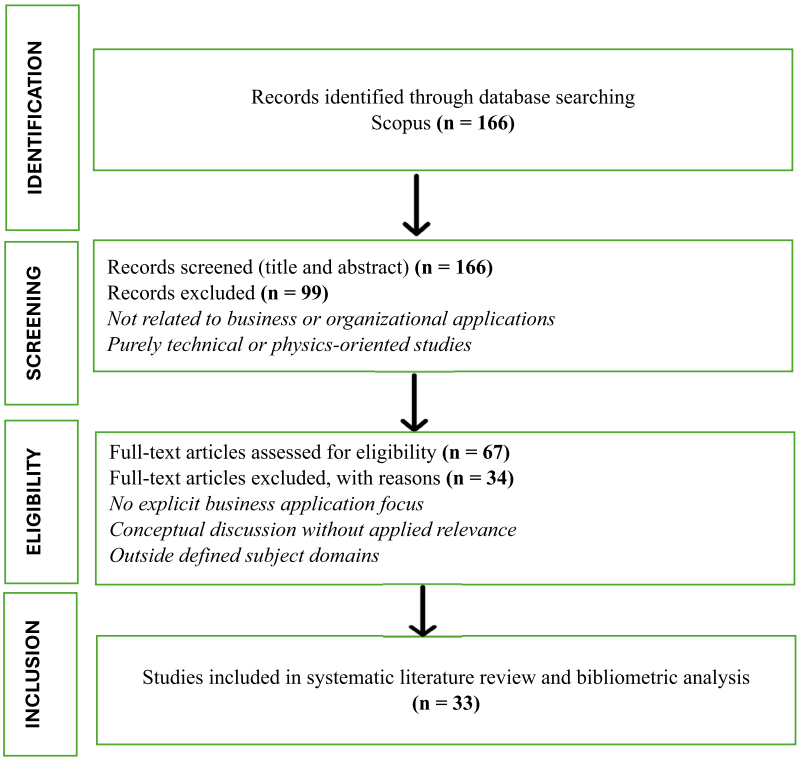

Figure 1 Selection Procedure Flowchart Using PRISMA Protocol

Table 1 Annual Research Output

Year	Articles
2004	3
2005	2
2006	2
2007	5
2008	1
2009	1
2010	4
2011	4
2012	1
2013	6
2014	8
2015	0
2016	0
2017	0
2018	0
2019	0
2020	3
2021	6
2022	8
2023	5
2024	9
2025	1

of quantum. At the same time, it prompted researchers to pivot away from the questions and issues regarding quantum computing and begin focusing on avenues of application, management, and industry-based research. The apex of output in 2024 demonstrates the

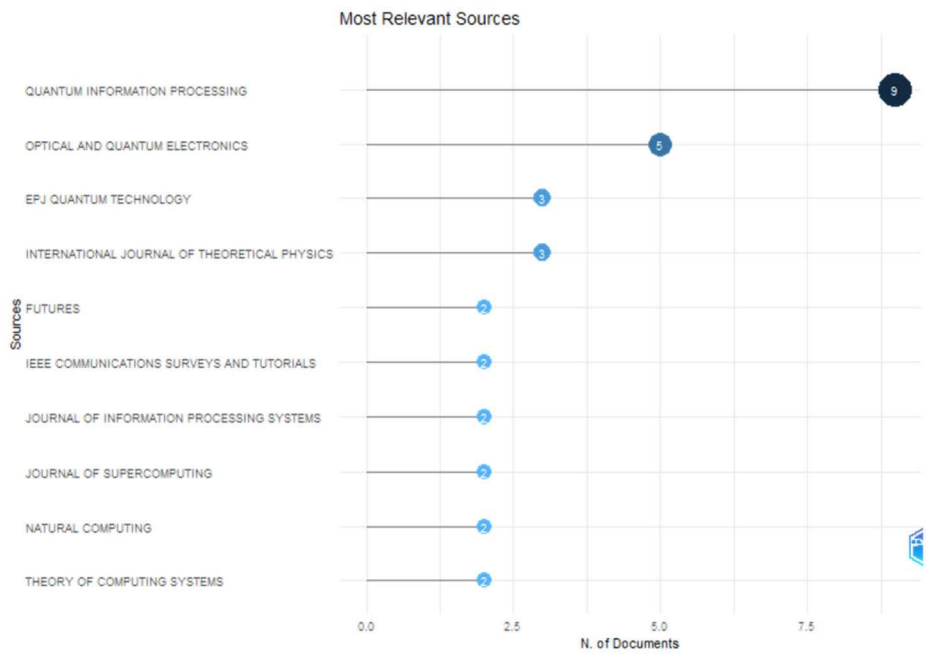

Figure 2 Most Pertinent Sources

strategic importance and rapid evolution of attention towards quantum computing research.

In total, the poly temporal records provided in Table 1 demonstrate increased scholarly engagement and validate the need to recognize and systematize studies on business applications, outcomes, and future directions associated with quantum computing as appropriate and relevant to this investigation.

In Fig. 2, the two most prolific sources of publication related to quantum computing literature depicted in the data set are displayed along with the most prolific category, overall. Not surprisingly, the data show a concentration of research in a thinly compiled number of specialized publication sources, signifying that quantum computing resides strongly in a technical, multidisciplinary output of research and literature surrounding the subject. Quantum Information Processing leads as the most prolific source of publication when research on quantum computing began to be published as a topic of scientific inquiry. Other prolific journal sources of publication include Optical and Quantum Electronics, EPJ Quantum Technology, International Journal of Theoretical Physics and other journals indicate that technology has rich roots in the quantum space studies versus hardware, quantum application, or computational theory.

In addition, the contributions referenced here from journals such as Futures and IEEE Communications Surveys and Tutorials, suggest a trend towards a more future focused, managerial and systems perspective, including extending implications for society and emerging communication technologies. Overall, the coverage here suggests a hybrid influence for quantum computing, with contributions coming from both the core technical genres, and a more expanded group of interdisciplinary sources, which would all contribute to the field evolving to more business, managerial and strategic applications.

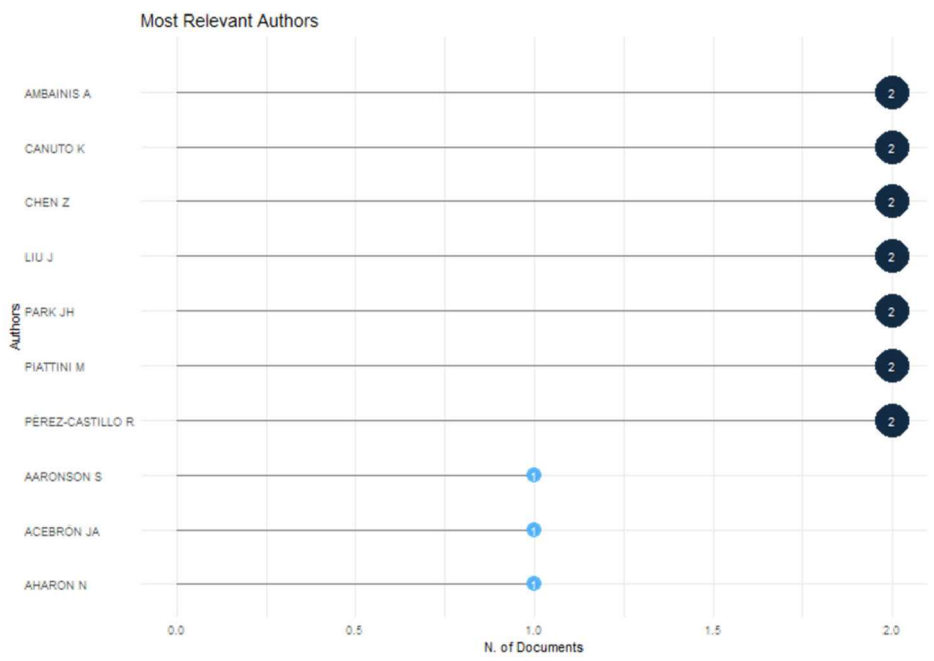

Figure 3 Most Pertinent Authors

Figure 3 identifies the most active and cited authors of quantum computing, based on the papers in this dataset. The pattern shows that several authors, such as Ambainis A., Canuto K., Chen Z., Liu J., Park J.H., Piattini M., and Perez-Castillo R., each had two contributions to the literature. Thus, there is an author distribution that is fairly balanced across these publications, without authorship for any specific author standing out more than others. This trend indicates the collaborative nature of quantum computing research, which is in an evolving field between multiple disciplines, in which advancement extends from theoretical physicists, computer scientists, and information systems to new business experiences in that space. Recognized contributors such as Aaronson S. and Aharon N. had fewer contributions in this survey, but they show the longstanding impact of these leading theorists and their subsequent influence on methodological and conceptual advances in the field, that are foundational to it.

The trend toward authorship fits existing trends into research ecosystems as convergent and collaboration is now the norm in the research space; suggesting that quantum computing is also being drawn from a more distributed research model despite not having concentrated clusters and approaches within disciplines.

Figure 4 highlights Important institutional affiliations in the quantum computing research ecosystem. Results reveal that both Information Engineering University and the Massachusetts Institute of Technology (MIT) are the most prolific institutions; each institution had six publications. They are successful due to the important role that quantum theory and cryptography have received worldwide and how advanced computational architectures informed and influenced by military research environments and high-end academic labs are driving these developments.

The Centre for Nanotechnology, Tel-Aviv University, and the Victorian Health Promotion Foundation also demonstrate considerable research activity, further supporting the

Figure 4 Most Pertinent Affiliations

concept of interdisciplinary expansion into new, emerging areas of quantum computing research related to nanoscience, theoretical physics, and health innovation. Contributions to quantum computing research from the Army Engineering University (Shijiazhuang), the Shenzhen Graduate School, and Universität zu Köln is also significant, demonstrating geographic diversification in quantum-related research with increased contributions from Asia and Europe.

The affiliation data suggests quantum computing research is driven by a mix of elite scientific institutions, and emerging global universities contributing to broader research ecosystem that may be supporting both fundamental and applied object-oriented research relevant to the research objectives of this study.

As can be seen in Fig. 5, ranking based on citation impact from quantum computing research, countries whose research has had significant impact were ranked in terms of citations received. Based on information gathered, China has gained an impressive lead with 353 citations received indicating a significant commitment by the nation toward advancing the use of quantum technologies. China also appears to demonstrate significant growth in both theoretical and practical applications of quantum sciences. India and the United States ranked second and third respectively with 191 and 160 citations indicating significant contributions toward the development of quantum algorithms, quantum hardware and implementation of quantum technologies and Interdisciplinary uses of quantum technologies within the finance, healthcare and cybersecurity Industries.

European countries including Germany, Latvia, Spain, and Sweden are demonstrating strong citation activity, which signifies continued support for collaborative research efforts through research networks both nationally across Europe and internationally.

The growth of countries such as Ukraine, Iran, and Georgia shows that there is a dissemination of research capability into the quantum arena and that not only are researchers

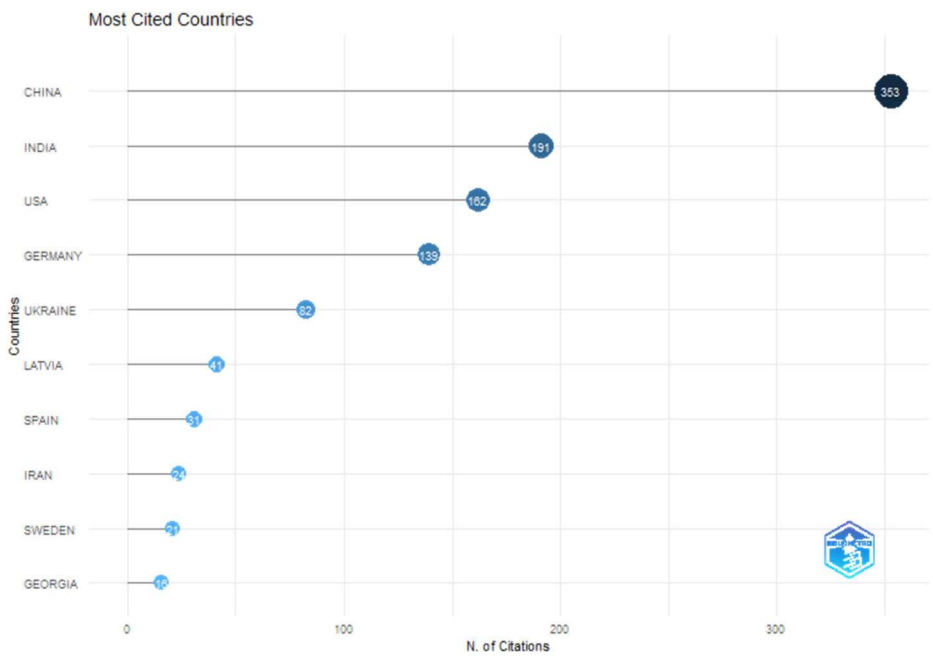

Figure 5 Most Cited Countries

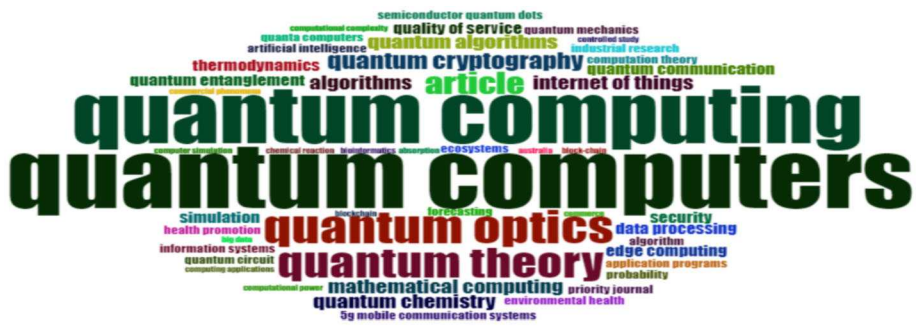

Figure 6 Word Cloud

from developing and transitional economies increasingly present, but also that having one such contributing presence should be considered to indicate an emerging science contributor to the global quantum computing community.

Based on the above citation data, it is indicated that research activity is showing geographical patterns of variances due to both established scientific research contributors and emerging contributor(s) to the international quantum computing community. Therefore, the significance of this study is reinforced to acknowledge global momentum for quantum technology development and the commercial utilization of this technology.

A word-cloud was created (Fig. 6) from the quantum computing literature in this dataset (n = 163) that illustrated the most commonly occurring terms in the collection of literature. The highest frequency of occurrence for the terms 'quantum computing', 'quantum computer', 'quantum optical', and 'quantum theory' further indicates the prevalence of research within the field, whether through theoretical or computational methods. In addi-

Table 2 Co-Word Net Details

Node	Cluster	Betweenness	Closeness	PageRank
mathematical computing	1	24.5480239	0.011627907	0.04505192
quantum chemistry	1	0	0.008474576	0.010789463
quantum entanglement	1	126.7899264	0.013513514	0.033015302
simulation	1	27.08236643	0.011494253	0.033063916
thermodynamics	1	45.13309631	0.010989011	0.029692407
algorithm	1	0	0.010416667	0.021420522
computation theory	1	0	0.007751938	0.007355507
forecasting	1	0	0.008403361	0.007270817
priority journal	1	2.548351648	0.009009009	0.026907292
probability	1	0	0.010416667	0.019632955
quantum mechanics	1	0.660526316	0.010526316	0.026203572
absorption	1	0	0.008403361	0.007270817
chemical reaction	1	0	0.008928571	0.020581203
commercial phenomena	1	0	0.008403361	0.007270817
controlled study	1	0	0.008403361	0.007270817
quality of service	2	38	0.007633588	0.023205897
5g mobile communication systems	2	0	0.00591716	0.009385718
industrial research	2	0	0.007575758	0.013276163
internet of things	3	108	0.010416667	0.029750463
block-chain	3	0	0.01010101	0.016119557
blockchain	3	0	0.01010101	0.016119557
quantum theory	4	214.8229646	0.013888889	0.047102691
algorithms	4	0	0.011235955	0.017605216
quantum algorithms	4	0	0.011235955	0.020484343
data processing	4	0	0.010869565	0.009249715
computer simulation	4	0	0.009090909	0.006715725
quantum computers	5	193.0155999	0.014705882	0.102850197
quantum computing	5	210.1481911	0.014705882	0.106717353
quantum optics	5	19.19061826	0.011904762	0.051517265
quantum cryptography	5	3.655764652	0.011494253	0.026282232
quantum communication	5	19.70699272	0.012195122	0.025099763
application programs	5	0	0.009615385	0.008803706
ecosystems	5	0	0.009708738	0.013958782
information systems	5	0	0.009708738	0.012572968
quanta computers	5	0.601214754	0.010309278	0.020842607
quantum circuit	5	0	0.00952381	0.008771982
big data	5	0	0.009615385	0.008803706
computational power	5	0	0.009708738	0.011305973
computing applications	5	0	0.009803922	0.014031544

tion, additional frequently occurring keywords of 'quantum cryptography', 'computational mathematics', 'thermodynamics', and 'algorithms' indicate the significant intersectionality of the current research within the fields of physics, computational science, ensured information security and applied algorithm development on multiple fronts.

The emergence of keywords such as 'Internet of Things', 'Simulation' and 'Artificial Intelligence' reflects the increasing layering of quantum technology on top of existing technological ecosystems and also an increasing relevance of quantum technology to business applications. Overall, these word-clouds provide a perspective of the breadth of theoretical implications and the multidisciplinary breadth of quantum computing literature. The word-cloud highlights the necessity of conducting a thorough and comprehensive literature review; thus, providing a thorough and comprehensive literature review is a necessity not only in order to cover the various areas of quantum computing, but also to cover the various areas of quantum computing and applied research.

Table 2 displays the characteristics of the co-word metric, Cluster membership, Betweenness Centrality, Closeness Centrality, and PageRank for the main 16 key terms from

the corpus examined on the subject of quantum computing. The clustering structure displays five thematic clusters that suggest a diverse but connected framework of research production. Cluster 1 captures core, scientific concepts including quantum chemistry, quantum entanglement, mathematical computing, and thermodynamics, indicating a viable, and common theoretical mechanism for the field of quantum. High betweenness scores for concepts quantum entanglement (126.78) and thermodynamics (45.33) are suggestive of these concepts being tertiary keywords across the continuing sub-domain of science.

Clusters 4 and 5 were the use-based terminology, with their approximately high levels of centrality ascribed to quantum theory, quantum algorithms, quantum optics, quantum computing, and quantum computers. Quantum theory (betweenness 214.82) and quantum computers (betweenness 193.02) form the dominant hubs in the network indicating a connection of theoretical development to emerging business applications and technological development. The other core applied terminology, internet of things, data processing, and quantum cryptography-based centralities, present a nominally moderate centrality indicative of possible growth between quantum computing and a growing technological application.

For each of the measures, whether betweenness, closeness, or page rank, quantum computing, or quantum computers appear as high page rank scores potentially signalling structural worthiness or, at the very least, collision density, suggesting close-nestled co-occurrence, across multiple clusters into thematic areas. The co-word network demonstrate an evolving, or recruited, research area marked by relative balance between complex theoretical constructs and exponential application-related development.

Overall, this level of detail supports the study's reconceptualizing purpose, charting (1) contexts and spheres of the premiere research in quantum computing, and (2) possible academic constructs and concepts poignantly identifying the intersections between a developing academic system and its associated business-driven application space.

In the following Fig. 7, a thematic map displays the dominant research themes of quantum computing on the basis of centrality (relevance in the field) and density (extent of development). The theme mapping is a useful visual representation illustrating how the intellectual structure has evolved in this field and where research energy is currently focused.

The top right quadrant (Motor Themes) shows the higher density and centrality of well-developed themes such as mathematical computing, quantum chemistry, quantum communication, quantum optics, quantum computing, and quantum computers. This represents the scientific and technological core of the field and supports ongoing development and influence on the wider research landscape. Its positioning reinforces the marked significance of important quantum computing architectures and essential quantum communication systems, comprising the foundation of the theoretical portfolio and more recently, new business applications.

The top left quadrant (Niche Themes) presents specialized and somewhat well-developed themes such as data processing, quantum circuits, and bioinformatics. Niche Themes exhibit relatively high internal cohesion with low centrality, highlighting that themes are still performing specific domain innovations to inform the development of applied quantum computing. The topics do not yet exhibit strong centrality, but their level of sophistication suggests there is strong potential for further interdisciplinary growth of

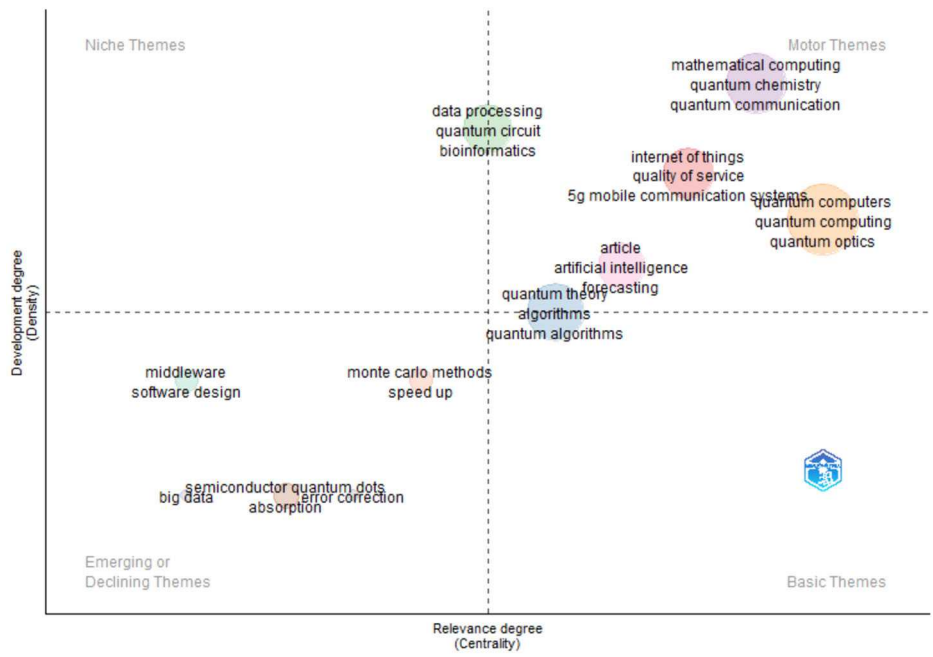

Figure 7 Thematic Map

applied quantum computing particularly in domains such as: computational biology and high-performance analytics.

In the bottom right quadrant (Basic Themes), keywords such as internet of things, quality of service, 5G mobile communication, artificial intelligence, forecasting and quantum algorithms are evident. These topics should be considered basic themes, foundational to the application-based growth of quantum computing, despite a lower degree of density. Their presence further attests to the fact that quantum computing is being increasingly adopted as a part of the digital transformation agenda more broadly, through networks of the IoT, AI models and next-gen communication infrastructures.

The lower left quadrant entitled Emerging or Declining Themes presents the slightly less mature topics of middleware, software design, big data, semiconductor quantum dots, and error correction. Some of the lower-left categorization could be viewed as declining interest in research, yet some of these topics of research, quantum dots and error correction, indicate a new area of development as opposed to a diminishing interest in research, still showing performance improvements similar to the performance improvements witnessed within the previous technology themes that stress issues of hardware stability, coherence, and scalability.

The thematic map presents a relatively healthy research ecosystem including older quantum bedrocks and newer professional applications of interdisciplinary basis supporting the rationale of this study as a systematic investigation of previous research pathways and new research pathways that include business applications demanding further exploration.

7 Discussion

7.1 Interpreting key findings: from theoretical potential to business reality

Finding 1: the NISQ–fault-tolerant distinction is the dominant determinant of near-term business feasibility Although the literature reviewed has consistently classified quantum applications by their computational depth, error tolerance, and qubit requirements, variational quantum eigen solvers (VQEs), quantum approximate optimizations of problems (QAOPs), hybrid quantum machine learning (QML) models, and sampling-based tasks have been specifically developed to work under NISQ-era hardware limitations [3, 6]; specifically, they are based on shallow (few gates) circuits, error mitigation techniques, and optimization loops (classical solutions) which keep their use technically possible with only hundreds of noisy qubits at the moment. Alternatively, applications such as Shor's algorithm (which has a cryptographic significance), simulating quantum chemistry with an accuracy of 100%, and large-scale QML require millions of logical qubits with full error correction [11, 12]. This means that, based on our synthesis of what's available, these later types of applications cannot currently be supported by existing or expected NISQ hardware. This technical distinction becomes especially strategic because it allows organizations to differentiate between those who engage in experimentation compatible with NISQ devices and those who plan for a disruptive future dependent upon fault tolerance; otherwise, resources could easily be wasted developing experimental methodologies with NISQ technology, while preparing for up to three or more years until those applications can be successfully deployed.

Thus, as organizations look to invest in the near-term future, their investments must be framed as the development of a development capability which generates knowledge, rather than as new applications being ready for implementation. Investing in fault-tolerant applications demands that organizations take necessary preparatory steps (as previously described) related to cryptographic migration, workforce development, and research into deploying fault-tolerant algorithms.

Finding 2: quantum readiness is sector specific and structurally determined There are three observable factors that govern how prepared specific sectors will be for a transition to quantum computing. These are: (1) The similarity of the problem structure and the types of advantages expected from quantum computing. (2) The ability of the organisation to make digital advancements and have the skills necessary to implement quantum computing; and (3) The willingness of the organisation to willingly accept uncertainty in their experiments.

A number of examples exemplify these three factors. Cryptography and Cybersecurity will have immediate applications because businesses will ultimately need quantum computing to effectively utilise their cryptographically secure information. Businesses in the financial services sector will also be relatively prepared for a transition to quantum computing due to the potential benefits associated with optimised processes using NISQ algorithms, and because these businesses have adequate financial resources, technology and analytical skills to realise those benefits.

Businesses in other sectors including Supply Chain Optimisation and Quantum Enhanced Machine Learning will see a moderate level of readiness for transition to quantum computing. In some instances, we can theoretically see a significant benefit to the sector from using NISQ algorithms to solve combinatorial optimisation problems. However,

the classical solutions we currently have will still be valid and inexpensive to implement for now, so cost will not drive many companies to adopt quantum computing in the near future.

Healthcare, Pharmaceuticals and Material Science will likely not see significant quantum computing capabilities until a great deal of fault tolerant hardware is developed within those sectors. We can observe that the patterns will not allow quantum to be adopted uniformly across all sectors. Rather, the adoption of quantum must be customised to meet the needs of each sector based on the specific technological capabilities and current state of readiness for the organisation(s) involved in that sector.

Finding 3: hybrid quantum classical architectures are the only viable implementation pathway Pure quantum solutions were not found in the real world of business by any of the studies. Instead, pure quantum solutions are virtually non-existent in real-world business applications. Hybrid quantum-classical architectures, where quantum processors are used for small defined tasks and integrated into classical workflows dominate. Variational methods are a clear example of this, where quantum circuits produce candidate solutions and classical methods adjust the parameters.

Hybrid System Design Challenges (such as the latency of the quantum-classical interface, orchestration complexities and co-optimisations) are not currently being addressed in the same way as pure quantum systems. This knowledge suggests that further progress into quantum computing will move from solving distinct problems using just a quantum algorithm towards preparing the quantum classically hybrid system engineering as a middleware between the two.

7.2 Positioning against existing literature and theory

This work supports the conclusions of previous domain-specific reviews relating quantum computing to finance, logistics and healthcare [9, 22, 27], and established technical evaluations of the limitations of NISQ technology [25]. The major contribution of this work is to extend these findings beyond technical viability and address the organizational and strategic theory implications.

Through application of the Technology-Organization-Environment (TOE) model, the evidence indicates that barriers to quantum adoption occur mainly at the organization/environment levels than at a future technological level. The complementary analysis via Resource Based View (RBV) confirms that at present, quantum computing capabilities display partially VRIN characteristics; they are potentially valuable and rare; but, due to NISQ status, they do not meet the definitions of inimitability or non-substitutability. As such, these findings clarify the reasons for the current outcome of sustained experimentation without any significant commercial deployment and on-going industry's development activities.

To provide a theoretically based solution to the disparate adoption paths observed across organizations, we introduced the Quantum Computing Strategic Alignment (QCSA) framework to map quantum adoption as an issue of strategic alignment; where issues of technology fit, organizational capability, and strategic value are the areas of consideration for successfully integrating quantum computing into the operation of an organization, as opposed to it being only a direct outcome of the availability of technology. The QCSA

framework integrates technological impediments and organizational readiness while providing a structured rationale to assist in the business-oriented decision-making process in relation to algorithm-driven feasibility assessments.

Framework structure The framework is structured along two primary analytical dimensions, Technological Readiness and Organizational Capability, with Strategic Value incorporated as an evaluative overlay. This structure allows quantum applications to be positioned comparatively, highlighting alignment or misalignment between feasibility, capability, and value creation potential.

Technological readiness The extent to which a quantum application can be performed using hardware that is currently available or hardware that can reasonably be expected to be developed in the near future is denoted as the "technological readiness" of that quantum application.

In terms of determining whether or not a quantum application is technically feasible, such a determination is made based upon the following factors: the required number of gates to implement a circuit (i.e., circuit depth); the degree of noise immunity; the number of physical qubits required; and whether or not error correction is used within the quantum circuit framework. The analysis results in two types of application-type groupings.

The first type includes all NISQ-compatible applications that use either error mitigation methods (e.g., variational quantum eigensolvers, QAOA-based optimizations, quantum samples, hybrid quantum machine learning models) or classical postprocessing methods to reduce errors resulting from using deep quantum circuits. This category of application types can be executed with the currently available hardware or hardware that may be available in the near-term.

The second category includes all applications that require error correction methods and require deep quantum circuits that use large numbers of physical qubits. Therefore, examples of such implementation include any of the known implementation of the cryptographically relevant versions of Shor's algorithm, quantum chemical simulations of a high degree of accuracy, and scalable quantum models of artificial intelligence. There is currently no known set of hardware available today or hardware that can reasonably be expected to be developed in the near future that can perform such applications. From a technological readiness perspective, applications in the first category (NISQ-compatible) are bound by a feasibility constraint that is more stringent than the second class of applications.

Organizational capability The absorptive and integrative capacity of organisations to test, consolidate and derive benefit from quantum technologies governs how effectively firms will utilise quantum technology within the organisation. By comparing a set of literature dictionaries, it is evident that the majority of organisations are successful in developing organisational capabilities that enable organisations to effectively adopt quantum technologies based on assessment of the overall organisational capability versus having direct access to quantum hardware alone.

The organisations' overall capability is comprised of four major components that consistently appear throughout the literature; (1) The organisation has a sufficient level of human capital that covers a broad range of disciplines including, but not limited to, quantum algorithm development, software engineering and formulating specific domain problems in

the context of quantum algorithms. (2) A sufficiently developed digital infrastructure and hybrid infrastructure with the addition of quantum classical integration (high end classical computing and quantum classical integration). (3) Sufficient financial resources (financial slack) and a level of risk tolerance (financial risk tolerance) that will support a long-term commitment to testing and developing new opportunities and innovations that present a significant level of uncertainty. (4) An organisational learning culture that enables organisations to develop and implement a sustainable governance framework to protect and support exploratory projects that may not provide any immediate returns.

The organisations that do not have these sets of capabilities and resources will encounter many challenges and obstacles (both structural and otherwise) regardless of the technological feasibility of using quantum technology, therefore demonstrating the need for organisations to align internal capabilities with external opportunities when using quantum technologies.

Strategic value Strategic value captures the business relevance of potential quantum advantage and is incorporated as an evaluative overlay within the framework. The synthesis indicates that strategic value is shaped by three interrelated factors: the magnitude of potential computational advantage relative to state of the art classical approaches, the scope of business impact in terms of cost reduction, revenue generation, risk mitigation, or regulatory compliance, and temporal urgency, particularly where delayed adoption generates asymmetric risk, as observed in post quantum cryptography.

Treating strategic value as an overlay allows applications with similar feasibility and capability profiles to be differentiated based on their broader organizational and competitive significance.

Strategic positions The four strategic positions derived from positioning applications in the QCSA framework include the following four areas:

- **The Immediate Action Zone** contains applications that have NISQ compatibility, which is being targeted by organizations with high capabilities, as opposed to companies with only moderate capabilities, that also have a very high or critical strategic value (e.g., financial optimization and post-quantum cryptographic transition planning).

- **The Capability Accumulation Zone** consists of fault tolerant dependent applications that are viewed as having very high long-term strategic value but that are not technologically ready yet (e.g., pharmaceutical drug discovery and advanced material simulation). While the investment to develop these applications is justified, there are other applications that would generate a more immediate return on investment.

- **The Structured Experimentation Zone** includes applications with NISQ compatibility that are viewed as having moderate strategic value, and with medium capability within the organization. Applications in this zone are identified as having an opportunity for learning that outweighs the potential of immediate performance application (e.g., logistics routing, energy optimization, and exploratory quantum machine learning).

- **The Observation Zone** consists of applications that are fault tolerant dependent applications and exhibit low organizational capability and uncertain strategic value. An example of such applications would be speculative use cases of quantum artificial intelligence.

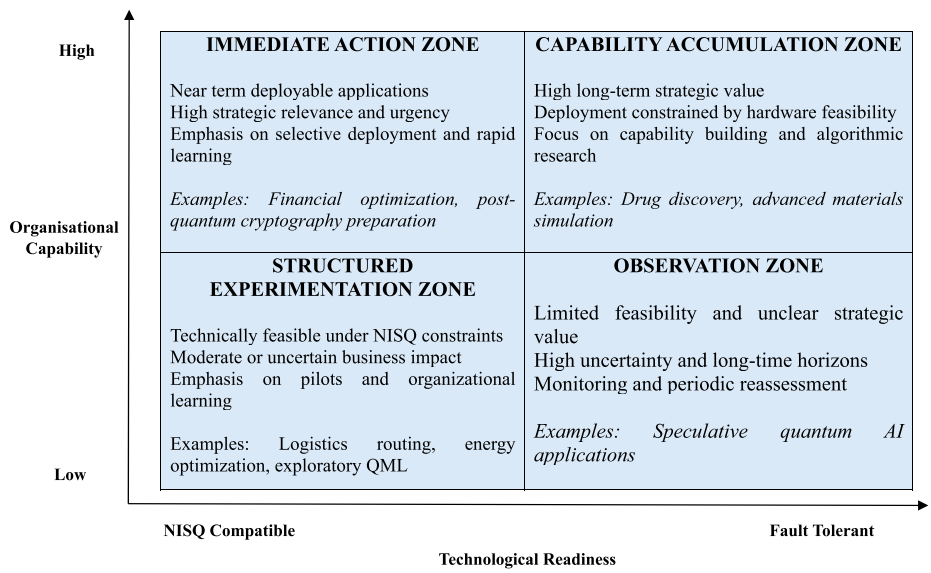

Figure 8 Quantum Computing Strategic Alignment (QCSA) Framework

Application of the framework In three steps, the QCSA analytical process classifies quantum applications by Technology Readiness, evaluates Organizational Capability Profiles on observable indicators for skills, infrastructure and risk tolerance, and evaluates Strategic Value overall through impact magnitude and timing. This process allows cross-sector comparisons based on their position on the Technology Readiness and Organizational Capability Matrix, the identification of misalignment between capability and feasibility, and the differentiation between Preparation, Experimentation and Deployment.

The Quantum Computing Strategic Alignment (QCSA) Framework is illustrated in Fig. 8 where it shows the positioning of quantum applications based upon Technology Readiness and Organizational Capability as well as Strategic Value.

8 Implications

8.1 Theoretical implications

The primary purpose of the research is to enhance the body of knowledge around Technology Adoption, Strategic Management, and Emerging Computing Systems by identifying key differences between quantum computing and incremental innovation in Digital Computing. The research also expands the Technology-Organization-Environment (TOE) Framework by showing that quantum computing demands a complete reinterpretation of the technology, organisation and environmental contexts. With respect to the technology, the research illustrates that the adoption of quantum computing is determined by the ability to distinguish between the NISQ era of quantum computing versus the Fault-Tolerant era. The research also highlights how the geographical and bi-political environment has a significant impact on the way organisations adopt the emerging technology.

With respect to Alternative Viewpoints, the Resource-Based View (RBV) has been enhanced by the research. By identifying that the capabilities associated with quantum computing will exhibit contingent VRIN characteristics and be time-dependent, it explains why there has been a continual cycle of experimentation with quantum computing to date

but no large scale competitive advantages. Furthermore, the research shows that the timing and ecosystem placement of quantum computing will be critical in transforming the emerging technology into a strategic resource.

The Quantum Computing Strategic Alignment (QCSA) Framework introduced in this research integrates the TOE and RBV perspectives to assess the triple bottom line value of quantum computing applications. The QCSA provides a methodical way to evaluate investments into emerging technologies with high uncertainty and long maturation cycles. In addition, the QCSA Framework provides a conceptual framework that may be applicable to other emerging paradigms such as quantum communications, quantum sensing, and neuromorphic computing.

By foregrounding hybrid quantum-classical systems, this research advances understanding of heterogeneous computer systems. Specifically, the research classifies quantum computing as a complementary capability that exists within hybrid classical computing systems rather than as a standalone device. The focus of this work is on issues such as the need for co-optimization, orchestration and the necessity for developing middleware technology.

8.2 Managerial implications

The findings highlight that Quantum computing is a long-term capability building initiative rather than a productivity technology that should be implemented right away. In the short term, an organization will focus on establishing a foundational quantum awareness, identifying business problems where quantum technology could be helpful, and having the opportunity to conduct low-risk experiments via hybrid quantum-classical workflows. Being prepared for post-quantum cryptography should be viewed as a separate, high-priority concern, meaning companies must take steps to prepare for this well in advance of the availability of quantum computers. Quantum computing systems take a long time to migrate to production systems and remain viable for many years following production.

In the middle term, organizations that have sustained interest in Quantum Computing should build the internal capability to develop hybrid architecture, conduct structured experimentation and have collaboration between the functional groups with the goal of making this activity organizationally relevant. As long as organizations involve themselves in the larger quantum computing ecosystem through collaborative research, standard-setting, and pre-competitive initiatives, they can further reduce the risk associated with developing capability

Longer term, Quantum Computing should be incorporated into an organization's enterprise architecture and into an organization's strategic planning processes. The governance structure and security policies, as well as the organization's culture, need to support an environment where collaboration occurs between different disciplines, uncertainty is tolerated, and learning continues to occur. Companies that take a realistic view of the technological timelines for quantum initiatives and have the capacity to absorb advances through fault tolerance will be in a better position to take advantage of the benefits of quantum computing, while simultaneously avoiding premature commitments.

There are widespread differences among sectors in terms of their stage of developing quantum computing capability, as it relates to the respective level of relevance to each of the sectors. Financial services and cybersecurity are experiencing the most immediate level of relevance, while healthcare and pharmaceuticals require longer-term pan-sector

preparations focusing on research partnerships and capability building. For both supply chain and logistics applications, organizations should consider selective experimentation while developing quantum computing capabilities in conjunction with ongoing improvements in classical optimization methods.

8.3 Policy and regulatory implications

The results from the work highlight the need for adaptive governance on a longer time horizon as quantum computing is still in a very long development period (and the potential for far-reaching systemic impact). Policymakers must continue investing in quantum research, common infrastructure, and building a skilled workforce given the high level of uncertainty and long timelines associated with commercialisation. They must also address the continuing shortage of highly skilled personnel and support the integration of interdisciplinary skills when developing educational and training programs. Governments must also provide coordination to establish a quantum ecosystem and to create the proper conditions needed for success through Development of Standards, Supporting Infrastructure and Coordination Between Private and Public Entities.

Regulatory frameworks should consider how best to mitigate the risk of quantum-related threats to information security, particularly the challenges posed by the development of new types of cryptographic systems. Policies designed to facilitate the transition to Post-Quantum Cryptography will help mitigate systemic vulnerabilities and ensure that the transition is carried out in an orderly manner. Export Controls and Technology Security Regulations must balance the need to protect national security with the need to maintain and support International Scientific Collaboration.

The rapid advancement of quantum technologies and the growing capabilities of quantum systems will require increased attention to the ethical, sustainability and equity implications of quantum technologies. Therefore, policy interventions at the local, national and international levels should be taken to promote responsible quantum technology development by establishing Ethical Governance Mechanisms, Conducting an Environmental Impact Assessment of Quantum Infrastructure, and Implementing Mechanisms to Prevent Cumulative Effects and Over-Concentration of Quantum Technology Capabilities. Furthermore, as the demand for quantum technologies continues to increase, there will be a continuing need for International Coordination in the Standardisation of Quantum Technology, the Development of Post-Quantum Cryptography and the Development of Responsible Technology Governance.

9 Conclusion

This study analyzes quantum computing from an organizational standpoint, offering a comprehensive investigation into the factors that will confer value to organizations at various stages of quantum technology development. Through the integration of systematic synthesis and bibliometric analysis, established organizational theories will be advanced alongside current discourse on quantum computing, aiming to formulate a more pragmatic strategic perspective on the potential of quantum computing, as opposed to a purely technology-centric assessment of its disruptive capabilities for the organizations. The study's findings unequivocally demonstrate that the significance of quantum computing for an organization is not contingent upon the creation of a finite number of algorithms

or technological advancements. Instead, the relevance of quantum technologies for an organization will fluctuate considerably depending on the degree of alignment among technological feasibility, organizational capabilities, and strategic intent as quantum technologies progress and the demand for quantum-based solutions emerges. To optimally leverage quantum technologies, organizations must consistently evaluate their competencies and establish and execute hybrid decision-making and strategic planning methodologies. This strategy will allow them to circumvent forming erroneous expectations regarding the disruptive potential of the quantum technologies. This study's conclusions create a foundation for more research and the creation of an evidence-based paradigm for developing quantum implementations within organizational infrastructures.

10 Future research agenda

Future research must explicitly explore real-world case studies to empirically validate how quantum computing is implemented, governed, and produces business value, rather than solely rely on simulating and conceptually demonstrating quantum computing for example. Longitudinal studies will enable researchers to capture how firms learn and adopt quantum technologies over time and how the forms of value realized from quantum technologies change as they become more broadly adopted. Standardized benchmarking frameworks must be developed to facilitate credible and transparent comparisons between quantum and classical computing solutions based on not only their computational performance, but their hardware development costs, execution time, and total cost of ownership.

Given the expectation that quantum computers will mostly be deployed as part of hybrid systems for some time, future research must focus on how organisations will engineer hybrid systems as opposed to focusing on the algorithmic performance of quantum computers. Researchers should investigate the association of optimal ways of decomposing computation across quantum and classical components, as well as how organisations will develop middleware and orchestrate workflows in hybrid systems. The co-optimisation of quantum and classical computing will become an essential research area in the quest to improve the performance of end-to-end solutions built on top of hybrid systems and researchers should seek to identify and reproduce reusable design patterns that incorporate insights from the software engineering, distributed systems, and enterprise architecture disciplines.

Research should also have a higher degree of sector specificity, including the development and experimental evaluation of class of quantum algorithms aligned to industry-specific problems and use cases in the sectors of finance, healthcare, supply chain management, and materials science, while considering NISQ era constraints and performing rigorous comparative analyses against leading classical solutions in the same domain. Quantum machine learning has notable importance in this regard because it will need to be empirically studied to understand the performance conditions, data requirements, generalisation behaviour, robustness, and limitations of quantum machine learning relative to rapidly evolving machine learning solutions being developed on classical systems.

From an organisational standpoint, there is an abundance of empirical research available to determine what factors influence organisations' decisions to adopt quantum computing based on size of firm and/or institutional context of organisations. Empirical research will also identify how organisations evaluate quantum investment opportunities, manage uncertainty in technology, and integrate quantum computing with their broader innovation

agendas. In terms of the workforce, further research must continue to develop workforce-related topics such as how to define skill requirements for quantum computing, create interdisciplinary education and training paradigms for quantum computing and identify talent management techniques that can breed the necessary skill set to meet the demand on a continuous basis. There is also a necessity to conduct complimentary total cost of ownership studies using integration costs and opportunity costs, as well as the need for organisations to change their business processes, to support the decision-making process of managers.

Research needs to focus extensively on how the adoption of quantum computing will impact the policy, ethical, and societal landscape. Researchers must place empirical focus on post-quantum cryptography as well as the challenges organisations face during their transition to post-quantum computing and the need for adequately designed regulations to assure a smooth transition. The broader societal impact of quantum computing, including digital inequality, economic concentration, workforce disruption, and international competition, warrant systematic investigation. Environmental considerations are also growing in prominence and therefore will require additional research, including a complete life cycle analysis to compare the environmental impacts of quantum computing with classical computing, in order to provide adequate support for responsible technology development and the governance of quantum technology.

11 Limitations

This study has limitations. The primary limitation of this study is its reliance solely on published, peer-reviewed, English-language journal articles from a defined timeframe; consequently, it overlooked other pertinent information sources, particularly those indicative of early-stage commercial or applied product development. The second limitation is that, owing to the exceedingly swift advancements in Quantum Computing Technology and Regulatory Frameworks, the conclusions drawn in this study are indicative of a specific temporal context and may not encompass all contemporary technological developments or nascent applications within this domain. A third weakness of this study is its reliance on secondary data, which precluded the evaluation of actual performance outcomes, the organizational adoption of products/services, or the return on investment. Consequently, as the discipline evolves, periodic updates and supplementary empirical research will be essential.

Author contributions
Sarin, G. wrote the manuscript. Sai R Y. and Sarin, M. updated the manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Use of AI tools declaration
GenAI tool Quillbot premium was used to improve the sentence structure and flow. Majority of the work was written by authors and given significant human touch.

Competing interests
The authors declare no competing interests.

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References

1. Acharya R, Aleiner I, Allen R, Andersen TI, Ansmann M, Arute F, Arya K, Asfaw A, Atalaya J, Babbush R, Bacon D, Bardin JC, Basso J, Bengtsson A, Boixo S, Bortoli G, Bourassa A, Bovaird J, Brill L, Broughton M, Buckley BB, Buell DA, Burger T, Burkett B, Bushnell N, Campero J, Chang H-S, Chen Z, Chiaro B, Chik D, Cogan J, Collins R, Conner P, Courtney W, Crook AL, Curtin B, Debroy DM, Del Toro Barba A, Demura S, Dunsworth A, Eppens D, Erickson C, Faoro L, Farhi E, Fatemi R, Flores Burgos L, Forati E, Fowler AG, Foxen B, Garcia G, Genois É, Giang W, Gidney C, Gilboa D, Giustina M, Gosula R, Dau AG, Gross JA, Habegger S, Hamilton MC, Hansen M, Harrigan MP, Harrington SD, Heu P, Higgott C. Suppressing quantum errors by scaling a surface code logical qubit. Nature. 2023;614(7949):676–81. https://doi.org/10.1038/s41586-022-05434-1.
2. Arute F, Arya K, Babbush R, Bacon D, Bardin JC, Barends R, Biswas R, Boixo S, Brandao FGSL, Buell DA, Burkett B, Chen Y, Chen Z, Chiaro B, Collins R, Courtney W, Dunsworth A, Farhi E, Foxen B, Fowler A, Gidney C, Giustina M, Graff R, Guerin K, Habegger S, Harrigan MP, Hartmann MJ, Ho A, Hoffmann M, Huang T, Humble TS, Isakov SV, Jeffrey E, Jiang Z, Kafri D, Kechedzhi K, Kelly J, Klimov PV, Knysh S, Korotkov A, Kostritsa F, Landhuis D, Lindmark M, Lucero E, Lyakh D, Mandrà S, McClean JR, McEwen M, Megrant A, Mi X, Michielsen K, Mohseni M, Mutus J, Naaman O, Neeley M, Neill C, Niu MY, Ostby E, Petukhov A, Platt JC, Quintana C, Rieffel EG, Roushan P, Rubin NC, Sank D, Satzinger KJ, Smelyanskiy V, Sung KJ, Trevithick MD, Vainsencher A, Villalonga B, White T, Yao ZJ, Yeh P, Zalcman A, Neven H, Martinis JM. Quantum supremacy using a programmable superconducting processor. Nature. 2019;574(7779):505–10. https://doi.org/10.1038/s41586-019-1666-5.
3. Bharti K, Cervera-Lierta A, Kyaw TH, Haug T, Alperin-Lea S, Anand A, Degroote M, Heimonen H, Kottmann JS, Menke T, Mok W-K, Sim S, Kwek L-C, Aspuru-Guzik A. Noisy intermediate-scale quantum algorithms. Rev Mod Phys. 2022;94(1):015004. https://doi.org/10.1103/RevModPhys.94.015004.
4. Cao Y, Romero J, Olson JP, Degroote M, Johnson PD, Kieferová M, Kivlichan ID, Menke T, Peropadre B, Sawaya NPD, Sim S, Veis L, Aspuru-Guzik A. Quantum chemistry in the age of quantum computing. Chem Rev. 2019;119(19):10856–915. https://doi.org/10.1021/acs.chemrev.8b00803.
5. Cerezo M, Arrasmith A, Babbush R, Benjamin SC, Endo S, Fujii K, McClean JR, Mitarai K, Yuan X, Cincio L, Coles PJ. Variational quantum algorithms. Nat Rev Phys. 2021;3(9):625–44. https://doi.org/10.1038/s42254-021-00348-9.
6. Cerezo M, Larocca M, García-Martín D, Diaz NL, Braccia P, Fontana E, Rudolph MS, Bermejo P, Ijaz A, Thanasilp S, Lumbreras-Zarapico E, Tazhigulov RN, Wierichs D, Sharma K, Dborin J, Sanchez V, Coles PJ. Does provable absence of barren plateaus imply classical simulability? Or, why we need to rethink variational quantum computing. 2023. arXiv preprint arXiv:2312.09121. https://doi.org/10.48550/arXiv.2312.09121.
7. Chen L, Chen L, Jordan S, Liu Y-K, Moody D, Peralta R, Perlner R, Smith-Tone D. Report on post-quantum cryptography (NIST IR 8105). National Institute of Standards and Technology; 2016. https://doi.org/10.6028/NIST.IR.8105.
8. Egger DJ, Gambella C, Marecek J, McFaddin S, Mevissen M, Raymond R, Simonotto A, Woerner S, Yndurain E. Quantum computing for finance: state-of-the-art and future prospects. IEEE Trans Quantum Eng. 2020;1:1–24. https://doi.org/10.1109/TQE.2020.3030314.
9. Feld S, Roch C, Gabor T, Seidel C, Neukart F, Galter I, Mauerer W, Linnhoff-Popien C. A hybrid solution method for the capacitated vehicle routing problem using a quantum annealer. Front ICT. 2019;6:13. https://doi.org/10.3389/fict.2019.00013.
10. Feynman RP. Simulating physics with computers. Int J Theor Phys. 1982;21(6–7):467–88. https://doi.org/10.1007/BF02650179.
11. Fowler AG, Mariantoni M, Martinis JM, Cleland AN. Surface codes: towards practical large-scale quantum computation. Phys Rev A. 2012;86(3):032324. https://doi.org/10.1103/PhysRevA.86.032324.
12. Gidney C, Ekerå M. How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits. Quantum. 2021;5:433. https://doi.org/10.22331/q-2021-04-15-433.
13. Grover LK. A fast quantum mechanical algorithm for database search. In: Proceedings of the 28th annual ACM symposium on theory of computing. ACM; 1996. p. 212–9. https://doi.org/10.1145/237814.237866.
14. Herman D, Googin C, Liu X, Galda A, Safro I, Sun Y, Pistoia M, Alexeev Y. A survey of quantum computing for finance. Nat Rev Phys. 2023;5(8):450–65. https://doi.org/10.1038/s42254-023-00603-1.
15. Humble TS, Thapliyal H, Munoz-Coreas E, Mohiyaddin FA, Bennink RS. Quantum computing circuits and devices. IEEE Des Test Comput. 2019;36(3):69–94. https://doi.org/10.1109/MDAT.2019.2907130.
16. Kjaergaard M, Schwartz ME, Braumüller J, Krantz P, Wang JI-J, Gustavsson S, Oliver WD. Superconducting qubits: current state of play. Annu Rev Condens Matter Phys. 2020;11:369–95. https://doi.org/10.1146/annurev-conmatphys-031119-050605.
17. Leymann F, Barzen J. The bitter truth about gate-based quantum algorithms in the NISQ era. Quantum Sci Technol. 2020;5(4):044007. https://doi.org/10.1088/2058-9565/abae7d.
18. Mays N, Pope C, Popay J. Systematically reviewing qualitative and quantitative evidence to inform management and policy making in the health field. J Health Serv Res Policy. 2005;10(Suppl 1):6–20. https://doi.org/10.1258/1355819054308576.
19. Montanaro A. Quantum algorithms: an overview. npj Quantum Inf. 2016;2(1):15023. https://doi.org/10.1038/npjqi.2015.23.
20. Mosca M, Piani M. Quantum threat timeline research report 2024. evolutionQ & Global Risk Institute; 2024. https://www.evolutionq.com/publications/quantum-threat-timeline-research-report-2024.
21. Nielsen MA, Chuang IL. Quantum computation and quantum information. 10th anniversary ed. Cambridge: Cambridge University Press; 2010. https://doi.org/10.1017/CBO9780511976667.
22. Orús R, Mugel S, Lizaso E. Quantum computing for finance: overview and prospects. Rev Phys. 2019;4:100028. https://doi.org/10.1016/j.revip.2019.100028.

23. Outeiral C, Strahm M, Shi J, Morris GM, Benjamin SC, Deane CM. The prospects of quantum computing in computational molecular biology. Wiley Interdiscip Rev Comput Mol Sci. 2021;11(1):e1481. https://doi.org/10.1002/wcms.1481.
24. Paul J, Criado AR. The art of writing literature review: what do we know and what do we need to know? Int Bus Rev. 2020;29(4):101717. https://doi.org/10.1016/j.ibusrev.2020.101717.
25. Preskill J. Quantum computing in the NISQ era and beyond. Quantum. 2018;2:79. https://doi.org/10.22331/q-2018-08-06-79.
26. Shor PW. Algorithms for quantum computation: discrete logarithms and factoring. In: Proceedings 35th annual symposium on foundations of computer science. IEEE; 1994. p. 124–34. https://doi.org/10.1109/SFCS.1994.365700.
27. Zinner M, Dahlhausen F, Boehme P, Ehlers J, Bieske L, Fehring L. Quantum computing's potential for drug discovery: early stage industry dynamics. Drug Discov Today. 2021;26(7):1680–8. https://doi.org/10.1016/j.drudis.2021.06.003.

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